

Dexter Fernandes

Bristol • +44 07436007369 • dexterfernandes11@gmail.com • linkedin.com/in/dexterferns

Machine Vision Engineer

Machine Vision Engineer with 3.5 years of C++ and Python expertise delivering deterministic, real-time vision solutions across oil & gas, construction and industrial safety applications. Proven track record integrating IR and RGB camera streams on Intel edge platforms under strict memory, timing and environmental constraints. Skilled in cross-functional collaboration, working closely with hardware, software and certification teams. Passionate about embedded AI and real-time vision, I thrive on building impactful technology that directly contributes to driver safety.

WORK EXPERIENCE

Esbaar • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Developed time-series machine learning models for the classification of on-shore oil rig operations to increase worker safety and reduce supervision required during 24/7 operation, achieving a 23% higher F1 score than human experts.
- Built an automated sensor-data cleansing pipeline (3.6 M data points) that cut model roll out from 3 days to less than 24 hours.
- Spearheaded the end-to-end development and production rollout of YOLOv5 models, to elevate rig worker and equipment detection precision by 15% in 3 months.
- Refactored a legacy monolithic system into a Kafka-driven microservice architecture, containerising each service with Docker to enhance failure isolation and improve MTBF by 35% within four months
- Enhanced code quality and documentation standards by leading formal peer code reviews and quality-assurance activities within each sprint cycle, ensuring best practices.

LivNSense Technologies Pvt. Ltd. • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI

- Led a 4-engineer team on ViCAS to design, train, and deploy deep learning models for real-time object detection, pose estimation, and tracking on edge hardware to reduce false positives by 33% while achieving sub-50 ms inference latency within 6 months.
- Integrated NCm-261-R Smart IR camera and implemented end-to-end C++ inference pipelines with the OpenVINO SDK on Intel edge devices using adaptive low-light preprocessing to ensure deterministic, real-time performance under strict memory and timing constraints.
- Deployed a unified fixed-weight model for IR and RGB streams on VTC7252-7C4IP Intel Core i7 edge computers, using FP16 quantization via OpenVINO to achieve deterministic, real-time inference.
- Implemented rigorous testing and validation frameworks based on simulated adverse conditions, to ensure model accuracy and stability, reducing performance variance by 25% under challenging lighting, occlusion, and noise scenarios.
- Accelerated embedded vision inference by redesigning the inference pipeline and model optimisation techniques to boost FPS by 233% (4.5 FPS to 15 FPS) to enable real-time identification and reporting of OSHA safety violations.
- Engineered synthetic training data pipelines using OpenCV and NumPy, reducing edge-case model development time from 6 days to 2 days for high-risk production environments.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

- Deployed real-time LiDAR SLAM (ROS2 Nav2 + Velodyne) on Jetson AGX Xavier, containerised with Docker for field-deployable, point-cloud mapping.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

CERTIFICATIONS

Deep Learning Specialisation

deeplearning.ai

Jul 2019

SKILLS

- **Programming Languages:** C++, Python
- **Perception:** GStreamer, LiDAR SLAM, Nav2, OpenCV, OpenVINO, ROS2, Vision Transformer (ViT), YOLOv5/v8
- **Backend & Cloud / Edge:** Apache Kafka, AWS, FastAPI, Intel VPU, Microsoft Azure, NVIDIA Jetson AGX, PostgreSQL
- **DevOps & MLOps:** Docker, DVC (Data Version Control), Git, Linux, MLFlow, Optuna
- **Edge Platforms & Hardware:** Intel Core i7 (VTC7252-7C4IP), Movidius Myriad VPU, NVIDIA Jetson AGX Xavier